

# Malak Ait Tamlihat

Ph.D. Candidate in High Energy Physics, Mohammed V University in Rabat

Member of the ATLAS Collaboration, CERN

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## Research Interests

Precision Collider Measurements, Detector Reconstruction and Performance, Machine Learning for Particle Physics, Collider Phenomenology, Monte Carlo Generation, Scientific Computing

## Education

**2021–Sep. 2026** **Ph.D. in High Energy Physics**, Faculty of Sciences, Mohammed V University, Rabat

*Thesis:* New physics searches in light-by-light scattering with the ATLAS detector and development of a production database for the HL-LHC ATLAS High Granularity Timing Detector.

Supervisor: Prof. Yahya Tayalati

**2018–2020** **M.Sc. in Matter and Radiation**, Faculty of Sciences, Mohammed V University, Rabat

Graduated first in the graduating class.

**2015–2018** **B.Sc. in Theoretical Physics**, Sultan Moulay Slimane University, Beni Mellal

Graduated first in the graduating class.

**2015** **Baccalaureate in Physical Sciences**, Beni Mellal

Graduated first in class.

## Research Experience

**2023–Present: ATLAS physics analysis: Measurement of light-by-light scattering in ultra-peripheral heavy-ion collisions with ATLAS Run 3 data, Searches for physics beyond the Standard Model: ALPs and massive gravitons**

I proposed a new interpretation idea for graviton searches and currently serve as the main analysis contact and lead analyzer for this ATLAS heavy-ion analysis. I coordinate and co-supervise the analysis, overseeing both the physics strategy and technical implementation. I prepared and produced the Monte Carlo samples required for the analysis, directed the development of the analysis software, and contributed to detector calibration and performance studies. I designed and implemented the statistical framework used to extract limits on axion-like particles and massive gravitons. I chair analysis meetings and lead the preparation of results for conference presentations and journal publications.

**2023-2026: Phenomenological papers**

**Main Author of two papers: JHEP 07 (2026) 001, PLB 846 (2023), 138237**

I proposed the initial research ideas and worked closely with my collaborators to develop them into comprehensive phenomenological studies. I implemented theoretical models in Monte Carlo event generators, generated Monte Carlo event samples, carried out the phenomenological analyses, produced publication-quality figures, validated the results, and contributed to writing the manuscripts.

**2022: ATLAS Heavy Higgs Search (Photon + MET): ATLAS publication note**

I participated in the development of the analysis workflow to reinterpret the results in new physics scenarios, produced dedicated samples, and performed statistical limit extraction. This work enabled the reuse of existing ATLAS results to constrain additional models, thereby strengthening the physics reach of the original analysis.

**2021–2022: Qualification Authorship project with ATLAS collaboration: technical work**

This project, carried out as my Qualification Authorship project within the ATLAS Collaboration, involved developing the initial MySQL production database and web tools for managing High-Granularity Timing Detector(

HGTD) component data. This infrastructure provides detector production tracking, quality-control tools, and centralized data management for HGTD production activities within the ATLAS Collaboration.

**Selected Publications, Complete publication list:** <https://inspirehep.net/authors/2643459>

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## Journal Articles

- **M. Ait Tamlihat**, G. Ez-Zobayr, L. Schoeffel and Y. Tayalati, "Searches for Extra-Dimensional Excitations in Light-by-Light Scattering," *JHEP* **07** (2026) 001. arXiv:2601.03110 [hep-ph].
- D. d'Enterria, **M. Ait Tamlihat**, L. Schoeffel, H.-S. Shao and Y. Tayalati, "Collider Constraints on Massive Gravitons Coupling to Photons," *Phys. Lett. B* **846** (2023) 138237. arXiv:2306.15558 [hep-ph].

## Conference Proceedings

- **M. Ait Tamlihat** et al., "Collider Constraints on Massive Gravitons," *Springer Proceedings in Physics* **425** (2025), 147–150. doi:10.1007/978-3-031-88933-2\_24.
- **M. Ait Tamlihat** for the ATLAS Collaboration, "Observation of Top-Quark Pair Production in Heavy-Ion Collisions with the ATLAS Detector," *PoS EPS-HEP2025* (2026) 196. doi:10.22323/1.485.0196.

## ATLAS Notes

- **ATLAS Collaboration**, "Measurement of light-by-light scattering and searches for axion-like particles and massive gravitons in ultra-peripheral Pb+Pb collisions at  $\sqrt{s_{NN}} = 5.36$  TeV." Internal Note: <https://cds.cern.ch/record/2919034>
- **ATLAS Collaboration**, "Search for High-Mass Resonances in Photon plus Missing Transverse Momentum Signatures in  $pp$  Collisions at  $\sqrt{s} = 13$  TeV with the ATLAS Detector," ATL-PHYS-PUB-2023-003 (2023).

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## Conference Presentations

### Oral Presentations

- XIV International Conference on New Frontiers in Physics (ICNFP 2025), Crete, Greece. "*Recent Results from Heavy-Ion Collisions with ATLAS*".
- European Physical Society Conference on High Energy Physics (EPS-HEP 2025), Marseille, France. "*Observation of Top-Quark Pair Production in Heavy-Ion Collisions*".
- LHC Forward Physics Workshop 2024. "*Collider Constraints on Massive Gravitons*".
- African Conference on High Energy Physics (ACHEP 2023), Rabat, Morocco. "*Collider Constraints on Massive Gravitons*".
- International Applied Physics Meeting, Institute of Applied Physics, Mohammed VI Polytechnic University, Benguerir, Morocco. "*Light-by-Light Scattering and Searches for Exotic Particles with the ATLAS Detector*".

### Poster Presentations

- 10th Annual Large Hadron Collider Physics Conference (LHCP 2022). "*Developing a Production Database for the ATLAS High-Granularity Timing Detector*".

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## Scientific Schools, Workshops & Internships

- Graduate Instrumentation and Detector School (GRIDS 2025), TRIUMF, Canada (2025).
- Rabat IEEE NPSS Summer School (REISS 2024), Rabat, Morocco (2024).
- CNRS-Africa Computing in High Energy Physics School (CHACAL 2024), Johannesburg, South Africa (2024).
- Asia–Europe–Pacific School of High-Energy Physics (AEPSHEP 2022), Busan, South Korea (2022).
- 2nd EPS Technology and Interfaces Group Hands-on Workshop, CERN, Switzerland (2022).

- First Pan-African Astro-Particle and Collider Physics Workshop, Rabat, Morocco (2022).
- European School on Instrumentation for Particle Physics (ESIPAP 2021), France (2021).
- ESCAPE Summer School: Data Science for Astronomy, Astroparticle and Particle Physics, France (2021).
- **Internships at European Organization for Nuclear Research (CERN):** Participated in technical activities of the ATLAS experiment, including ATLAS detector shifts and High-Granularity Timing Detector (HGTD) test beam operations. (Aug–Oct, 2022, Nov–Jan, 2025)

## Teaching & Outreach

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- Lecturer, *Introduction to Particle Physics*, Master's program, Faculty of Sciences, Mohammed V University, Rabat (2025, 2026)
- Lecturer, *Advanced Linux*, Master's program, Faculty of Sciences, Mohammed V University, Rabat (2025)
- Instructor, *Monte Carlo and Geant4 Simulation Techniques* (practical sessions), Master's program, Faculty of Sciences, Mohammed V University, Rabat (2023–2024)
- Organizer, International Masterclass: *Hands-On Particle Physics* (2023)
- Co-organizer and instructor, ATLAS data-analysis tutorial, Institute of Applied Physics, Mohammed VI Polytechnic University (9–10 May 2023)
- Organizer of ATLAS/KM3NeT collaboration meetings, Faculty of Sciences, Rabat
- Member of the organizing committee, First Pan-African Astro-Particle and Collider Physics Workshop, Rabat (21–23 March 2022)
- Session Chair, *Astroparticle Physics and Cosmology*, African Conference on High Energy Physics (ACHEP 2023)
- Member of the Local Organizing Committee, African Conference on High Energy Physics (ACHEP 2023), Rabat, Morocco

## Honors & Awards

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- **Winner**, 18th L'Oréal–UNESCO *For Women in Science* Maghreb Young Talents Programme (2024). One of five laureates selected across the Maghreb in recognition of outstanding scientific excellence among early-career women researchers.
- **First Prize**, Morocco Youth Prize for Scientific Research (2024). National award presented by the Moroccan Ministry of Youth, Culture, and Communication to recognize outstanding scientific achievements by young researchers.

## Technical Skills

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**Programming Languages:** C++, Python, LaTeX

**Monte Carlo & Simulation:** MadGraph5\_aMC@NLO, STARlight, SuperChic, Geant4, detector simulation, Monte Carlo event generation

**HEP Software:** ROOT, PyROOT, ATLAS Analysis Framework, GitLab

**Web & Databases:** MySQL, PHP, HTML, CSS, JavaScript

**Operating Systems:** Linux, macOS, Windows

## Languages

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**Arabic:** Native

**English:** Fluent (C2)

**French:** Fluent (C2)